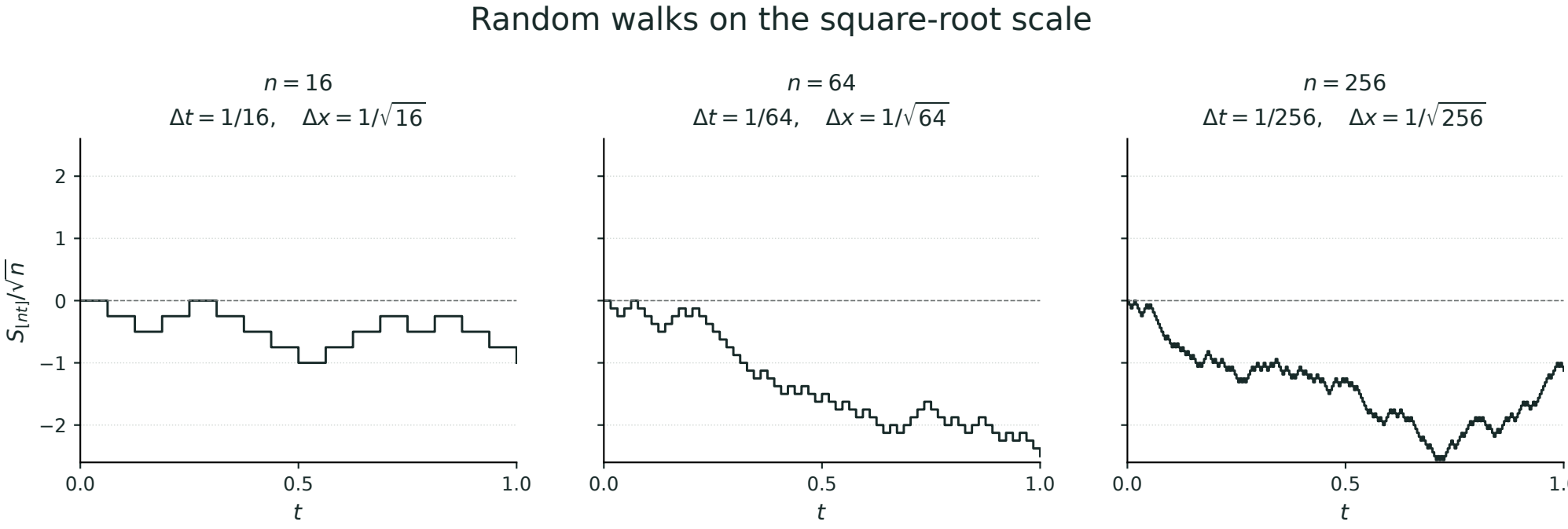
Random walks on the square-root scale
$n = 16$
$\Delta t = 1/16, \quad \Delta x = 1/\sqrt{16}$
$n = 64$
$\Delta t = 1/64, \quad \Delta x = 1/\sqrt{64}$
$n = 256$
$\Delta t = 1/256, \quad \Delta x = 1/\sqrt{256}$
$S_{\lfloor nt \rfloor}/\sqrt{n}$
2
1
0
−1
−2
0.0
0.5
1.0
t
Seed 918273 · Shared increment prefixes, separately rescaled; no pathwise refinement or limit proof.